

Robust Multimodal Framework (RMMF): Treating Modality Availability as a First-Class Input for Inference-Time Missing-Modality Robustness

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Abstract

Multimodal learning achieves strong performance by jointly exploiting heterogeneous data, yet virtually all existing architectures assume complete modality availability at inference time, an assumption that is routinely violated in practice. We introduce Robust Multi-Modal Framework (RMMF), a lightweight, backbone-agnostic framework that augments any multimodal backbone with three mechanisms. **(i) Interaction gating:** a binary modality mask encodes which inputs are present and gates the cross-modal interaction blocks accordingly, so a block whose query or key modality is absent is skipped instead of being fed zero-padded features, which also saves computation in proportion to the missing modalities. **(ii) Specialized prediction heads:** provide one dedicated classifier per modality-availability pattern (2^{K-1} heads under an indispensable-modality design), each learning a decision boundary tuned to its own modality subset rather than a single shared head that compromises across patterns. **(iii) Modality dropout training:** randomly removes optional modalities during training, so every head receives gradient signal and the shared encoder learns features that transfer across availability patterns. RMMF adds only a few $\times 10^4$ head parameters, from 0.24% of the Tri-Modal Survival Estimation (TMSE) backbone to 8.3% of the compact Fusion and Orthogonal Projection (FOP) backbone, and it lowers inference cost roughly in proportion to the absent modalities. We validate the approach across three settings in two unrelated domains, wrapping a different published backbone in each. In cancer survival prediction we wrap TMSE, a tri-modal survival model that fuses whole-slide images, genomic pathway profiles, and pathology reports (Concordance index on TCGA-Bladder Urothelial Carcinoma (BLCA): **0.687**→**0.747**, paired AR test **p=0.031**). In audio-visual speaker identification we wrap CoMISI, a model that fuses voice and face embeddings through cross-modal feature interaction (near-perfect accuracy ($\geq 99.96\%$) when the visual stream is present, with a controlled drop when it is absent). In cross-lingual multimodal speaker identification we wrap FOP (Fusion and Orthogonal Projection), which fuses face and audio embeddings under an orthogonality constraint that separates identities (**POLY-SIM** Grand Challenge 2026, RMMF on FOP: **0.734**→**0.962** overall in the inductive setting). This RMMF result is a core component of our 1st-place challenge entry, whose full system additionally applies transductive test-time label propagation to reach **0.999**. In all three settings RMMF matches or improves the full-modality result of the original backbone, with statistically significant survival gains on all three TCGA cohorts, and it degrades gracefully when modalities are missing, which makes missing-modality robustness a property obtainable by wrapping existing fusion archite

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1. Introduction

Multimodal systems combine heterogeneous data sources to exploit the complementarity of different information channels, consistently outperforming unimodal counterparts across a wide range of tasks [1, 2]. In computational pathology, jointly modeling histopathology slides, genomic profiles, and clinical reports captures both the morphological and molecular dimensions of disease, leading to substantially more accurate survival predictions [3]. In biometric systems, fusing acoustic and visual cues provides robustness against noise and illumination changes that each individual modality would fail to handle alone [4]. Despite this success, multimodal architectures share a critical vulnerability: they are designed under the assumption that all modalities are present at inference time. In clinical deployments, genomic profiling may not be ordered for a patient, pathology reports may be unavailable or inconsistently formatted. In speaker identification, cameras may be absent or occluded. In multilingual deployments, the test language may differ from training, breaking the statistical assumptions of the audio encoder. Ma et al. [5] systematically studied this vulnerability in multimodal transformers, showing that performance can degrade sharply when a single modality is missing at test time and that the best fusion strategy is dataset-dependent rather than universal.

Existing approaches each carry significant limitations. Imputation methods [6] train generative models to reconstruct absent representations, but the reconstructed features may be unreliable in high-stakes applications such as cancer prognosis. Knowledge distillation [7, 8] transfers full-modality representations to incomplete-modality students, but introduces architecture-specific teacher-student dependencies. Shared-encoder approaches [9, 10] learn embeddings robust to missing inputs by jointly training on multiple availability patterns, but provide no structured mechanism for exploiting whichever specific modality combination is available at inference. Modality dropout [11], which stochastically drops modalities during training to counter gradient imbalance, is closest to our work, however, it neither resolves how cross-modal computation should be routed when a modality is absent, nor provides dedicated decision boundaries for different availability patterns. We propose Robust Multi-Modal Framework (RMMF), which treats modality availability as an explicit, first-class input to the model through three complementary mechanisms: **Interaction gating**, any cross-modal block whose query or key modality is absent is skipped entirely, preventing the model from attending over zero-padded representations and saving computation proportional to the number of absent modalities. **Specialized prediction heads**, a dedicated head is instantiated for each distinct modality availability pattern. Each head learns the optimal combination strategy for its specific modality subset without interference from other patterns, and is selected automatically at inference based on the observed mask. **Modality dropout training**,

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each optional modality is independently dropped with probability p_{drop} during training, ensuring every head receives adequate gradient signal and shared encoder layers develop representations useful across availability patterns. A central design principle is the **indispensable modality**, one modality per application is designated as always available (Whole Slide Images (WSI) in pathology, audio in speaker recognition), reflecting domain knowledge about the primary data source and keeping the head count at 2^{K-1} for K total modalities. The main contributions of this work are:

- **A backbone-agnostic missing-modality framework.** RMMF wraps any existing multimodal backbone through three drop-in mechanisms, interaction gating, per-availability-pattern prediction heads, and modality dropout, with no change to the backbone’s internal blocks or its training objective. We instantiate the same wrapper on three unrelated backbones, Tri-Modal Survival Estimation (TMSE) for tri-modal cancer survival, CoMISI for audio-visual speaker identification, and Fusion and Orthogonal Projection (FOP) for cross-lingual face-voice identification, covering two domains and several modality types (imaging, genomics, text, audio, and video).
- **Interaction gating with a formal cost analysis.** Any cross-modal block whose query or key modality is absent is skipped rather than fed zero-padded inputs (Eq. (1)), so inference cost falls with each missing modality. We give a per-inference cost analysis as a function of the number of absent modalities (Section 3.3), and we measure both sides in practice: dropping the optional modality cuts a single forward pass to 26–46% of the full-modality cost, while the added prediction heads raise the parameter count by only 0.24% to 8.3% of the backbone (Table 1).
- **Per-availability-pattern specialized heads, with an inductive-bias argument.** RMMF instantiates one prediction head per modality-availability pattern, 2^{K-1} heads under an indispensable-modality design that fixes one always-present modality per application (the WSI in pathology, audio in speaker recognition) and halves the head count from 2^K to 2^{K-1} (Section 3.2). Section 3.4 gives the inductive-bias argument for this choice, showing why dedicated per-pattern decision boundaries dominate a single shared head once the availability patterns induce a distribution shift in the fused representation.
- **Cross-domain validation.** A single fusion principle is validated on cancer survival prediction and on (cross-lingual) audio-visual speaker identification, including a 1st-place result on the POLY-SIM 2026 Grand Challenge, where RMMF alone raises inductive P-accuracy from 0.734 to 0.962 and the full challenge system reaches 0.999. The same three mechanisms improve results across these unrelated domains and backbones without any per-domain modification, which suggests the benefit derives from the fusion design rather than the specifics of a single domain.
- **A consistent empirical finding:** RMMF matches or improves the full-modality result of the original backbone in all three settings, not only the missing-modality case. On TCGA survival the full-modality C-index rises from 0.687 to 0.747 on Bladder Urothelial Carcinoma (BLCA), with statistically significant gains on all three

cohorts (paired Approximate Randomization (AR) test, $p = 0.031$ on BLCA), which indicates that modality dropout acts as a beneficial regularizer on the shared encoder rather than merely a robustness patch.

The rest of this paper is organized as follows. Section 2 reviews related work. Section 3 details the RMMF framework, its complexity analysis, and the inductive-bias argument. Section 4 describes the three use-case instantiations. Section 5 reports experimental settings, results, ablations, and discussions. Section 6 discusses limitations and broader impact, Section 7 represents the reproducibility statement, and Section 8 concludes the paper.

2. Related Works

2.1. Multimodal Fusion and Representation Learning

The landscape of multimodal fusion is surveyed in depth by Baltrušaitis et al. [1], who categorize approaches along five axes: representation, translation, alignment, fusion, and co-learning. Among these, fusion has attracted the most research attention. Early and late fusion strategies are simple but either require strict modality alignment or fail to capture cross-modal feature interactions. Intermediate fusion addresses this by computing cross-modal interactions at the representation level, where cross-attention mechanisms enable one modality’s features to query another’s, capturing fine-grained inter-modal dependencies. Xu et al. [12] provide a detailed survey of transformer-based multimodal architectures, covering attention bottlenecks and modality-specific tokenization strategies. What is common to all these methods is the closed-world assumption of complete modality availability, which our work directly lifts.

2.2. Multimodal Survival Prediction in Computational Pathology

Survival analysis from WSI has been shaped by Multiple-Instance Learning (MIL), which aggregates patch-level features into slide-level predictions. CLAM [13] established an attention-based MIL baseline that remains widely used, TransMIL [14] extended this with a transformer backbone to capture spatial correlations between patches. More recent work addresses the computational burden of processing thousands of patch tokens, MHIM [15] mines hard instances through masking to focus attention on diagnostically relevant regions, while R2T-MIL [16] improves patch representation quality through foundation model re-embedding. For prototype-based slide representations, PANTHER [17] learns unsupervised morphological prototypes from WSI, providing compact and interpretable slide summaries. Multimodal fusion in pathology largely pairs WSI with genomic data. MCAT [18] introduced co-attention between WSI patches and gene expression pathway vectors to learn their joint interactions. Multimodal Optimal Transport-based Co-Attention Transformer (MOTCAT) [19] replaced co-attention with optimal transport to better align the two modalities’ distributions. CMTA [20] added cross-modal translation as an auxiliary task, forcing each modality’s encoder to predict the other’s features. MMP [21] extends prototype learning to jointly prototype WSI and genomic data. TMSE [3], our pathology backbone, introduces a three-modality system (WSI, genomics, and expert pathology reports) with a Context-aware Tissue Prototype (CTP) module combining differentiable Gaussian Mixture Model

(GMM) aggregation [22] with a Mamba [23] state-space model for long-range contextual modeling. Cross-modal alignment uses Attention-Entropy Interaction (AEI) modules regularized with matrix-based Rényi entropy [24] to prevent smaller modalities from being overwhelmed by the large WSI token sequence. Despite TMSE’s advances, it, like all prior methods, assumes complete modality availability at inference, a limitation our work directly addresses.

2.3. Multimodal Speaker Recognition

Speaker recognition has advanced from GMMs to deep embedding networks, with Emphasized Channel Attention, Propagation, and Aggregation in Time Delay Neural Network (ECAPA-TDNN) [25] having become one of the most widely adopted speaker embedding architectures. It combines channel attention, residual propagation, and multi-scale aggregation to produce speaker embeddings that generalize robustly across conditions. Incorporating the visual modality provides complementary information, particularly when acoustic features degrade in noise or reverberation. Early audio-visual systems fused independently trained models at the score level. Shon et al. [26] improved this with attention-based fusion, dynamically emphasizing the more reliable modality. A comprehensive review of AI-based speaker identification is provided by Jahangir et al. [27]. CoMISI [4], our speaker recognition backbone, advances beyond attention-based fusion with the Cross-Modal Interaction Fusion (CMIF) module. CMIF projects the audio and visual embeddings into a shared 128-d space (linear layer, batch normalization, and Rectified Linear Unit (ReLU)) and applies a learnable per-feature emphasis (importance scaling) to each. It then distills their cross-modal relation by passing the element-wise product of the two emphasized embeddings through a fully connected layer, and adds the resulting relation vector back to both modality embeddings to enrich each with cross-modal context before concatenation and classification. CoMISI achieves state-of-the-art accuracy on Global Research Identifier Database (GRID) and Ryerson Audio-Visual Database of Emotional Speech and Song (RAVDESS) but, like all prior work, has no mechanism for handling visual modality absence. Terraf and Iraqi [28] separately studied robust feature extraction for speaker identification in diverse acoustic environments, providing useful context for the robustness challenges our framework addresses.

2.4. Robustness to Missing Modalities

Ma et al. [5] provided the first systematic study of missing-modality sensitivity in multimodal transformers, showing that degradation can be severe and that the optimal fusion strategy is dataset-dependent, findings that motivate a principled architectural solution. A comprehensive survey by Wu et al. [29] covers the full landscape, identifying imputation, distillation, and shared-encoder learning as the three dominant paradigms. Among shared-encoder approaches, ShaSpec [9] learns both shared (modality-invariant) and modality-specific features, enabling the model to fall back on the shared space when a modality is absent. MISA [10] enforces invariant and specific subspaces through distributional similarity and orthogonality objectives. However, neither provides a mechanism for exploiting the specific available modality combination, the prediction pathway is the same regardless of what is present. Li et al. [30] systematically investigated what architectural and training choices lead to robustness, finding no dominant

strategy across datasets and conditions, further motivating our structured, pattern-specific approach. Wang et al. [11] introduced modality dropout in the context of video classification to address gradient imbalance between modalities, showing that standard joint training can cause multimodal models to underperform their best unimodal counterpart. RMMF builds on modality dropout as a training technique but adds two structural components absent from prior work, interaction gating prevents meaningless cross-modal computations when a modality is absent, and specialized heads provide dedicated decision boundaries for each availability pattern, giving the model an explicit structured understanding of what is available rather than relying solely on implicit dropout-induced robustness.

3. RMMF: Method and Analysis

3.1. RMMF Framework

Figure 1 gives an overview of the RMMF framework. Let a multimodal system process K modalities with features $\mathbf{f}_1, \dots, \mathbf{f}_K \in \mathbb{R}^d$. Modality 1 is the indispensable modality (always available), modalities $2, \dots, K$ are optional.

3.1.1. Modality Mask and Feature Initialization

We define the **modality mask** $\mathbf{m} = (m_2, \dots, m_K) \in \{0, 1\}^{K-1}$, where $m_j = 1$ iff modality j is present. When $m_j = 0$, we set $\mathbf{f}_j = \mathbf{0}$. This is a deliberate no-op, because all cross-modal interactions involving absent modalities are gated (Section 3.1.2), the zero representation never reaches the prediction. The mask is computed automatically from the data pipeline at both training and inference time, requiring no manual intervention.

3.1.2. Interaction Gating

Let $\mathcal{B}_{ij}$ denote the cross-modal interaction block between modalities i and j . RMMF applies the following gating rule:

$$\text{Gated}(\mathcal{B}_{1j}) = \begin{cases} \mathcal{B}_{1j}(\mathbf{f}_1, \mathbf{f}_j) & \text{if } m_j = 1, \\ \mathbf{f}_1 & \text{otherwise (pass-through),} \end{cases} \quad (1)$$

where $j \in \{2, \dots, K\}$ indexes the optional modalities and $\mathbf{f}_1$ denotes the feature of the indispensable modality. When optional modality j is absent ($m_j = 0$), the block is skipped and the indispensable modality feature $\mathbf{f}_1$ passes through unchanged. The pass-through preserves output dimensionality regardless of mask value, ensuring full architectural compatibility with any backbone without internal restructuring.

3.1.3. Multi-Head Specialized Classifier

For each of the 2^{K-1} possible mask values $\mathbf{m}$, RMMF instantiates a dedicated prediction head $H_{\mathbf{m}}$:

$$\hat{y} = H_{\mathbf{m}}(\mathbf{z}_{\mathbf{m}}), \quad (2)$$

where $\mathbf{z}_{\mathbf{m}}$ is the mean-pooled concatenation of available post-fusion features. At inference, the appropriate head is selected based on the observed mask. Each head is a single linear layer. The motivation is straightforward, the optimal

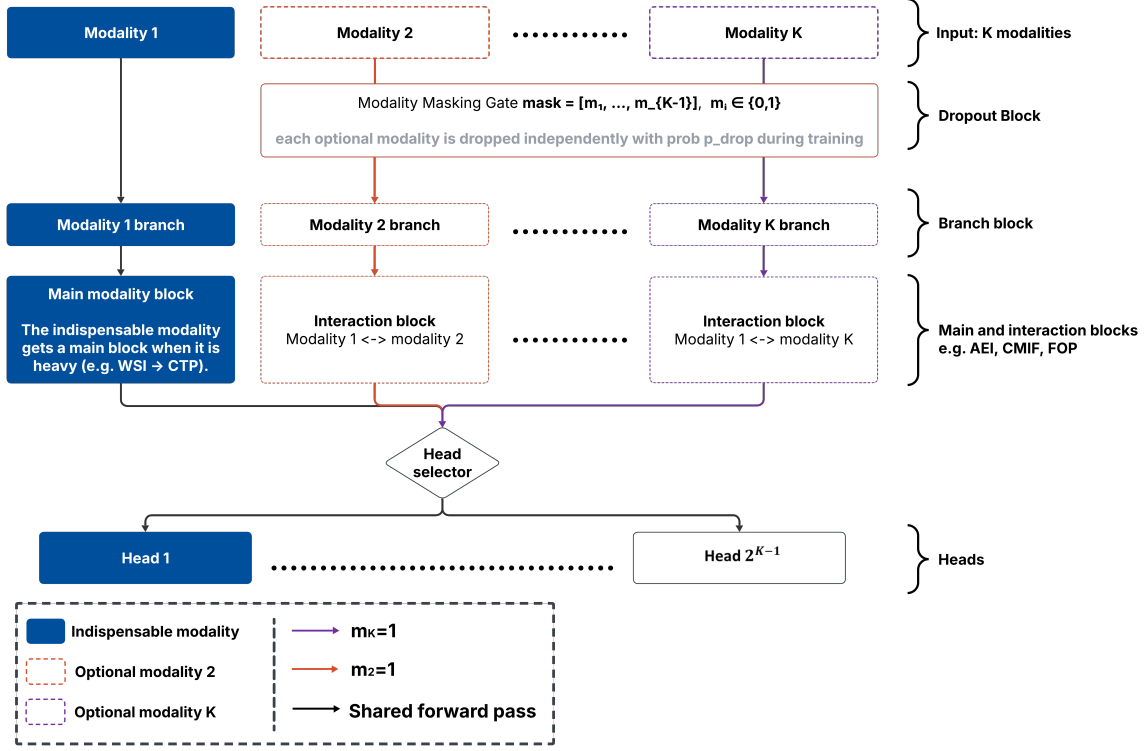


Figure 1: **The RMMF framework.** One modality is indispensable (solid blue) and always present, while the remaining modalities are optional (dashed) and gated by a binary mask. When a modality is absent its cross-modal interaction block is skipped. A main-modality block is added only when the indispensable modality is heavy, as for the WSI. The mask selects one of 2^{K-1} specialized heads, and modality dropout during training gives every head gradient signal for its availability pattern.

decision boundary for a model with access to all three modalities is genuinely different from the optimal boundary for a model with only WSI and genomics, or only WSI alone. A shared head trained across all availability patterns must compromise between them, dedicated heads eliminate this constraint. For $K = 3$ (one indispensable modality), this yields four heads, representing under 0.3% of the TMSE backbone’s parameters (Table 1).

3.1.4. Modality Dropout Training

At each training step, each optional modality j is independently dropped with probability p_{drop} (0.2 for pathology, 0.5 for CoMISI, and 0.5 for POLY-SIM). The mask is set accordingly, gating is applied per Eq. (1), and the loss is computed through the corresponding head $H_{\mathbf{m}}$:

$$\mathcal{L} = \mathbb{E}_{\mathbf{m} \sim \text{Bernoulli}(1-p_{\text{drop}})^{K-1}} [\mathcal{L}_{\text{task}}(H_{\mathbf{m}}(\mathbf{z}_{\mathbf{m}}), y)], \quad (3)$$

where $\mathbf{m} \in \{0, 1\}^{K-1}$ is the modality mask, each component m_j drawn independently as $m_j \sim \text{Bernoulli}(1 - p_{\text{drop}})$ so optional modality j is dropped with probability p_{drop} , $H_{\mathbf{m}}$ is the specialized head corresponding to mask $\mathbf{m}$, $\mathbf{z}_{\mathbf{m}}$ is the mean-pooled post-fusion feature vector produced under mask $\mathbf{m}$, y is the ground-truth label, and $\mathcal{L}_{\text{task}}$ is the

task-specific loss: the Negative Log-Likelihood (NLL) survival loss [31] with 4 time bins for pathology, or cross-entropy for speaker identification. The expectation $\mathbb{E}_{\mathbf{m}}$ is taken over all mask values realized during training through modality dropout, ensuring every head $H_{\mathbf{m}}$ receives gradient signal proportional to its dropout frequency. The dropout probability p_{drop} balances the training frequency of each head against the need to maintain sufficient full-modality samples for effective backbone training.

3.2. The Indispensable-Modality Design Principle

Designating one modality as always-present (WSI in pathology, always acquired in standard care, audio in speaker recognition, the primary identity signal) is both a domain-knowledge statement and a complexity control. Without it, the head count would be $2^K - 1$ (every non-empty subset), and training would have to cover the absence of the most informative modality, demanding far more incomplete-modality samples. With it, only the 2^{K-1} patterns over the optional modalities need heads, and the strongest signal is guaranteed at every inference. The choice is therefore a design lever: a poor choice (designating a weak modality as indispensable) would both inflate degradation and waste head capacity.

3.3. Computational-Complexity Analysis

Let the backbone contain, for the indispensable modality, one cross-modal interaction block $\mathcal{B}_{1j}$ per optional modality $j \in \{2, \dots, K\}$, each with cost c_j (for attention-style AEI/CMIF blocks, $c_j = O(n_1 n_j d)$ in token counts n_1, n_j and width d). Under an availability mask $\mathbf{m}$, gating (Eq. (1)) executes only the blocks with $m_j = 1$. The per-inference interaction cost is therefore $C(\mathbf{m}) = \sum_{j:m_j=1} c_j$, so the saving relative to the full-modality forward pass is exactly $\sum_{j:m_j=0} c_j$, the cost of every block tied to an absent modality is removed, formalizing the “saving proportional to the number of absent modalities.” When the interaction blocks are comparable ($c_j \approx c$) and $a = |\{j : m_j = 0\}|$ modalities are absent, the saving is ac , and the interaction stage scales linearly in the number of present optional modalities ($K - 1 - a$) rather than the constant ($K - 1$). The indispensable-modality CTP/WSI path always runs, so gating accelerates precisely the cross-modal stage that would otherwise attend over zero-padded inputs.

Parameter overhead. The only parameters RMMF adds are the 2^{K-1} linear heads. A head over a concatenation of r post-fusion blocks of width w into C classes costs rwC parameters, summed over the 2^{K-1} patterns this is a small constant independent of backbone depth. For $K = 3$, w a few hundred, and C a handful of classes/bins, the heads total on the order of 10^4 – 10^5 parameters, so the overhead is a fraction of the backbone that depends on the backbone’s own size (Table 1). Concretely, the TMSE instantiation ($K = 3$, $w = 256$, $C = 4$, bias-free heads) uses four heads of $4,096 + 3,072 + 3,072 + 2,048 = 12,288$ weights, only 8,192 beyond the single full head a non-robust model would carry, well under 0.2% of the multi-million-parameter backbone. The same accounting gives 4,386 and 3,096 added parameters for CoMISI on GRID and RAVDESS (3.8% and 2.7%) and 35,910 for FOP (8.3%), the larger fractions come from these backbones being small, not from wider heads. Table 1 reports the measured overhead and the single-sample GPU wall-clock in the full- and reduced-modality regimes for all three.

Table 1: Measured RMMF overhead per backbone: added head parameters (absolute and as a fraction of the backbone) and single-sample GPU wall-clock in the full- and reduced-modality regimes. Dropping the optional modality cuts inference cost to 26–46% of the full pass.

UC	Backbone	Added params	Overhead	Full $\rightarrow$ reduced (ms)
UC1	TMSE	+8,192	+0.24%	3.06 $\rightarrow$ 0.81 (WSI-only, 26%)
UC2	CoMISI	+4,386/+3,096	+3.8%/+2.7%	0.35 $\rightarrow$ 0.15 ($\neg V$, 43%)
UC3	FOP	+35,910	+8.3%	0.43 $\rightarrow$ 0.20 ($\neg$ face, 46%)

3.4. Why Per-Pattern Heads Beat a Shared Head

Consider two availability patterns $\mathbf{m}, \mathbf{m}'$ whose post-fusion feature distributions $P(\mathbf{z}_{\mathbf{m}})$ and $P(\mathbf{z}_{\mathbf{m}'})$ differ, which they must, since $\mathbf{z}_{\mathbf{m}}$ and $\mathbf{z}_{\mathbf{m}'}$ are produced by different gated computation paths over different inputs. A single shared head H is one linear map applied to both, it minimizes the mixture risk $\pi_{\mathbf{m}}R_{\mathbf{m}}(H) + \pi_{\mathbf{m}'}R_{\mathbf{m}'}(H)$ and therefore sits at a compromise that is Bayes-optimal for neither pattern whenever the per-pattern optimal boundaries differ. Per-pattern heads $H_{\mathbf{m}}, H_{\mathbf{m}'}$ decouple the objective into $R_{\mathbf{m}}(H_{\mathbf{m}}) + R_{\mathbf{m}'}(H_{\mathbf{m}'})$, each minimized independently, so the attainable risk is no larger and is strictly smaller whenever the optimal boundaries differ. The cost is sample-splitting (each head sees only its pattern’s samples), which is exactly what modality-dropout training counteracts by routing a controlled fraction of every batch to each head. RMMF thus trades a benign capacity increase (a few linear layers) for the removal of an inter-pattern compromise, beneficial precisely when availability patterns induce distribution shift, the regime of interest.

4. RMMF Application Studies

4.1. Application 1: Cancer Survival Prediction

We used TMSE [3] as the backbone (Figure 2), it processes three modalities, *WSI*: patches are extracted at $20\times$ via CLAM [13] and encoded with PLIP [32], a pathology foundation model pre-trained with visual-language supervision. They are then compressed by the CTP module, which combines differentiable GMM aggregation [22], clustering similar patches into $C_w = 16$ prototypes to reduce redundancy, with a Mamba [23] state-space model that captures long-range contextual dependencies across the full patch sequence, modeling Tumor Microenvironment (TME) signals that local prototypes alone cannot represent. *Genomics*: bulk RNA-seq data is organized into biological pathway vectors [33] and encoded with SNN [34]. *Pathology reports*: free-text reports are structured by a chain-of-thought LLM pipeline and embedded with BiomedBERT [35]. Cross-modal fusion uses two AEI modules, each applying cross-attention regularized with matrix-based Rényi entropy [24] to prevent the smaller modalities from being overwhelmed by the large WSI token sequence. **RMMF Instantiation** - WSI is the indispensable modality, it is always acquired as part of standard clinical care. Genomics (m_G) and reports (m_R) are optional, giving $\mathbf{m} = (m_G, m_R) \in \{0, 1\}^2$. When $m_G = 0$ (resp. $m_R = 0$), the AEI block computing WSI–genomics (resp. WSI–report) interaction is skipped per

Eq. (1). Four specialized heads are instantiated, one for each mask pattern. **Dataset** - We use three TCGA [36] cohorts: BLCA ($n = 328$), Breast Invasive Carcinoma (BRCA, $n = 875$), and Lung Adenocarcinoma (LUAD, $n = 401$), each providing WSI, pathway-vectorized RNA-seq, and pathology reports. Target: overall survival. Metric: concordance index (C-index), 5-fold cross-validation.

Experimental settings - All models are trained for 40 epochs with batch size 1 and gradient accumulation using the Multi-scale Momentum Muon (M3) optimizer [37] (learning rate 5×10^{-5} , weight decay 10^{-5} , fast momentum $\beta_1 = 0.9$, second-moment $\beta_2 = 0.999$, slow momentum $\beta_3 = 0.9$, slow chunk size 10, Newton-Schulz steps 3) with a constant learning rate schedule. NLL survival loss [31] with 4 time bins, GMM prototype count $C_w = 16$, $p_{\text{drop}} = 0.2$ applied independently to genomics and reports. The learned baselines in Table 2 (MCAT, MOTCAT, CMTA, MMP, and the MIL methods) are re-run under our identical 5-fold split and preprocessing, so the comparison is controlled rather than quoted from the source papers. Statistical significance is assessed with a paired AR test (10,000 resamples over the five folds) between RMMF-TMSE and our re-run TMSE baseline.

4.2. Application 2: Audio-Visual Speaker Identification

We used CoMISI [4] as the backbone (Figure 3), ECAPA-TDNN [25] produces 192-d speaker embeddings from 500 ms audio segments. FaceNet [38] produces 512-d face embeddings from video frames detected by Multi-task Cascaded Convolutional Networks (MTCNN). Inside the CMIF module, each embedding is projected into a shared 128-d space (linear layer, batch normalization, and ReLU) and rescaled by a learnable per-feature emphasis, the element-wise product of the two emphasized embeddings is passed through a fully connected layer to form a 128-d relation vector that is added back to both embeddings. The two enhanced 128-d embeddings are then concatenated (256-d) and classified. **RMMF Instantiation** - Audio is the indispensable modality. Visual is optional ($m_V \in \{0, 1\}$). When $m_V = 1$, the fused 256-d representation is classified by the full head, when $m_V = 0$, the CMIF cross-modal interaction is bypassed, the audio embedding is projected on its own (128-d) and routed to a dedicated audio-only head $H_{(0)}$. **Datasets** - The **GRID** dataset provides recordings from 34 speakers (18M, 16F) totaling 34,000 neutral utterances. **RAVDESS** provides 24 professional actors performing emotional speech (neutral, calm, happy, sad, angry), with 1,440 utterances. To model realistic deployment conditions, both datasets are augmented with: Gaussian noise (SNR 0–15 dB), reverberation (RT60 0.2–0.6 s, simulated with Pyroomacoustics [39]), and visual distortions (Gaussian noise, blurring, color shifting, cropping, rotation). Data split: 70/15/15 with no speaker overlap.

Experimental settings - All models are trained for 100 epochs with batch size 32 using the Adam optimizer (learning rate 10^{-3} , dropout 0.5) with cross-entropy loss. $p_{\text{drop}} = 0.5$ applied to the visual modality.

4.3. Application 3: Cross-Lingual Multimodal Speaker Identification (POLY-SIM Grand Challenge 2026)

The POLY-SIM Grand Challenge 2026 targets closed-set speaker classification under two simultaneous real-world difficulties: the face modality may be entirely absent at test time, and the audio modality may be in a language unseen during training. Participants must deploy a single unified model across all conditions, no retraining per setting is

allowed. The challenge provides FOP as the official baseline backbone. FOP is a two-stream pipeline that extracts face and audio embeddings independently and fuses them through a lightweight module that (i) combines complementary cues from both modalities into a fused embedding and (ii) imposes orthogonality constraints on the two modality embeddings to encourage discriminative identity separation in the joint space. Unlike metric-learning formulations that rely on pairwise or triplet losses, FOP uses a classification objective with orthogonality regularization, making training more efficient and avoiding the computational complexity of hard negative mining. **RMMF Instantiation** - We apply RMMF on top of the FOP codebase [40], with audio indispensable and the face modality optional, yielding two specialized heads (Figure 4).¹ **Datasets** - The challenge uses the Multilingual Audio-Visual Celebrity dataset (MAV-Celeb) dataset [41], a multilingual audio-visual collection of celebrity speech clips covering multiple spoken languages. MAV-Celeb was introduced to study the language-independence of face-voice association and to benchmark cross-lingual speaker recognition, two properties that are directly exercised by the POLY-SIM evaluation protocol. The evaluation metric is P-accuracy at candidate set sizes $P \in \{3, 4, 5, 6\}$, measuring whether the system correctly identifies the true speaker among P candidates. Four settings are evaluated: P3 (multimodal, same language), P4 (audio-only, same language), P5 (multimodal, cross-lingual), and P6 (audio-only, cross-lingual), the overall score is their mean.

Experimental settings - The model is trained for up to 300 epochs with early stopping (patience determined on a held-out 10% local validation split) using the Adam optimizer (learning rate 10^{-3} , weight decay 10^{-5} , batch size 32) with CosineAnnealingLR scheduling and label smoothing of 0.05. Modality dropout is applied with $p_{\text{drop}} = 0.5$, dropping the face modality per sample during training. The joint audio-face embedding dimension is set to 512 with dropout probability 0.5.

5. Experiments and Discussion

5.1. Application 1: Cancer Survival Prediction

Table 2 compares RMMF-TMSE against TMSE and a representative set of baselines spanning unimodal, bimodal, and tri-modal methods.

In the full-modality setting, RMMF-TMSE reaches a 5-fold C-index of 0.747 ± 0.045 on BLCA, 0.744 on Breast Invasive Carcinoma (BRCA), and 0.691 on Lung Adenocarcinoma (LUAD). Our re-run TMSE baseline scores 0.667 on BLCA, close to the 0.687 reported by the original authors, so RMMF improves the BLCA C-index by +0.080 over that re-run baseline. A paired AR test confirms a significant full-modality gain on all three cohorts: BLCA $p = 0.031$, BRCA $p = 0.038$, and LUAD $p = 0.032$. These gains in the full-modality case arise from two concurrent effects: modality dropout regularizes the shared encoder by preventing over-reliance on any single modality, reducing

¹The 1st-place challenge entry combined RMMF on FOP with two additional components of the full system: a domain-adversarial term to encourage language-invariant audio embeddings, and transductive label propagation at inference. The RMMF mechanisms (gating, per-availability heads, modality dropout) are the contribution evaluated here, the full-system score is reported in Table 4 for completeness.

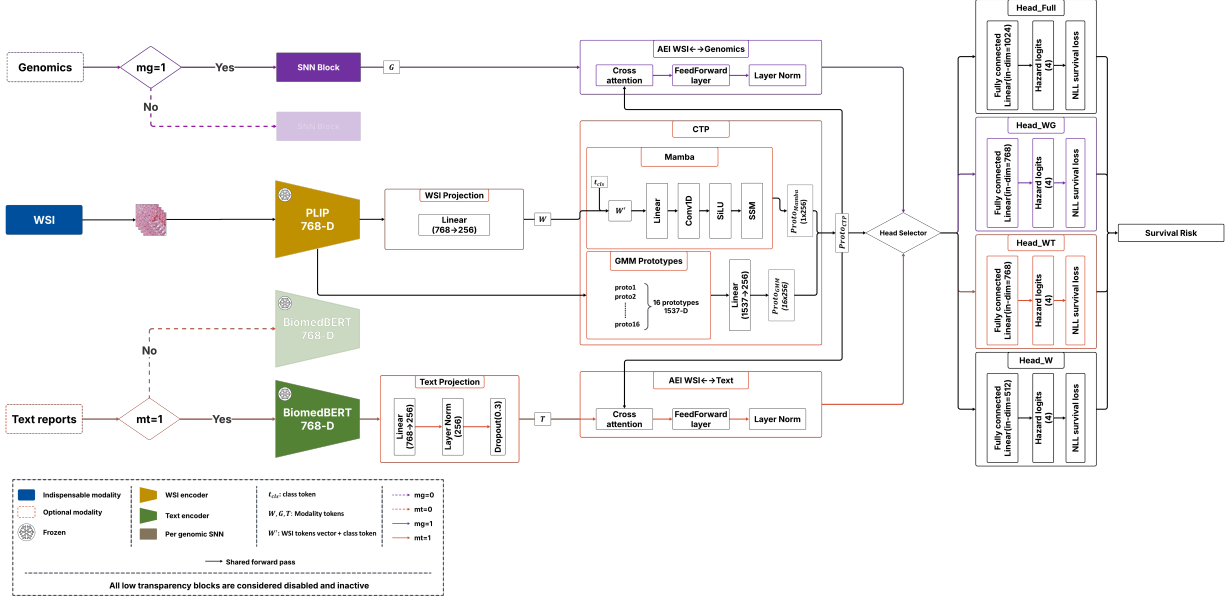


Figure 2: **RMMF on the TMSE backbone.** The WSI is the indispensable modality, encoded by PLIP and compressed by the CTP module that combines Gaussian prototypes with a Mamba branch. Genomic pathways and the pathology report are optional and each interacts with the WSI through an AEI cross-attention block. The mask (m_G, m_R) selects one of four heads, and every head predicts hazards over four survival bins trained with the NLL survival loss.

overfitting, and the specialized head $H_{(1,1)}$ learns a dedicated decision boundary for the complete feature distribution, without the interference of missing-modality training samples. Crucially, even when one or both optional modalities are absent, RMMF-TMSE maintains competitive performance. The WSI+genomics variant achieves 0.718 on BLCA, comfortably above the best bimodal baseline (MOTCAT, 0.673). The WSI-only variant stays competitive with dedicated WSI-only baselines (average 0.621, within the 0.595–0.627 span of those methods), even though WSI is only the fallback modality here rather than the design target. This confirms that the specialized heads trained under modality dropout learn genuinely robust representations, not degenerate fallbacks, because the shared encoder they rely on was trained with full-modality supervision.

5.2. Application 2: Audio-Visual Speaker Identification

Table 3 reports accuracy on all four evaluation conditions. RMMF-CoMISI reaches near-perfect accuracy ($\geq 99.96\%$) across all visual-present conditions, matching or exceeding the original CoMISI backbone.

The original CoMISI backbone is already at or near ceiling on three of the four conditions, leaving little full-modality headroom, the clearest improvement is on GRID under clean conditions, where RMMF-V raises accuracy from 0.9788 to 0.9996 (+2.08 pp, McNemar $p < 10^{-28}$ over seed-matched runs), while the remaining conditions are statistically indistinguishable from the backbone. We attribute the GRID-clean gain to the regularizing effect of modality dropout, by training the model to function without visual features, we prevent it from over-relying on visual

Table 2: C-Index (5-fold cross-validation mean) on TCGA survival prediction datasets. **Bold**: best result, underline: second best. G: genomics, W: WSI, T: pathology report.

Model	Mod.	BLCA	BRCA	LUAD	Avg.
SNN [34]	G	0.649	0.694	0.623	0.655
CLAM-MB [13]	W	0.655	0.604	0.594	0.618
TransMIL [14]	W	0.647	0.606	0.627	0.627
MHIM [15]	W	0.620	0.572	0.593	0.595
R2T-MIL [16]	W	0.651	0.605	0.611	0.622
PANTHER [17]	W	0.612	0.616	0.581	0.603
MCAT [18]	G+W	0.668	0.723	0.627	0.673
MOTCAT [19]	G+W	0.673	0.715	0.617	0.668
CMTA [20]	G+W	0.665	0.710	0.661	0.679
MMP [21]	G+W	0.674	0.716	0.607	0.666
TMSE [3]	G+W+T	<u>0.687</u>	<u>0.738</u>	<u>0.682</u>	<u>0.702</u>
RMMF-TMSE (W)	W	0.638	0.601	0.624	0.621
RMMF-TMSE (W+G)	G+W	0.718	0.724	0.671	0.704
RMMF-TMSE (W+T)	W+T	0.702	0.711	0.663	0.692
RMMF-TMSE (all)*	G+W+T	0.747	0.744	0.691	0.727

* Significant full-modality gain over the re-run TMSE baseline on all three cohorts

(paired AR test, 10,000 resamples: BLCA $p = 0.031$, BRCA $p = 0.038$, LUAD $p = 0.032$).

cues that are degraded or unreliable, which in turn improves performance even when visual features are present. When the visual modality is absent, performance degrades gracefully: the reduction relative to RMMF-V ranges from 1.16 pp on GRID clean to 6.30 pp on RAVDESS noisy. On GRID, where the audio alone is already highly speaker-discriminative, the audio-only head sits next to a standalone frozen-ECAPA-TDNN baseline (both near 99%). The gap is decisive on RAVDESS: the audio-only head reaches 95.1% (clean) while the frozen-feature baseline attains only 50.6%, because RMMF fine-tunes the audio encoder inside the shared model and recovers speaker identity that the frozen embeddings do not expose. This confirms that the audio-only head $H_{(0)}$ learns a genuine unimodal representation rather than a degenerate fallback.

Why the frozen audio features fail on RAVDESS. The 50.6% audio-only baseline on RAVDESS reflects a limit of the pretrained features, not of the missing-modality design. A capacity-free nearest-class-mean classifier on frozen ECAPA-TDNN embeddings already reaches 48.6%, and a trained linear probe on the same embeddings does no better, so the frozen features do not separate the RAVDESS speakers: their centroids are nearly collinear (mean inter-centroid cosine similarity 0.81, Fisher discriminant ratio 0.12). On GRID the same embeddings separate cleanly (inter-centroid cosine 0.15, nearest-class-mean accuracy 1.00), which is why the frozen baseline is already near ceiling

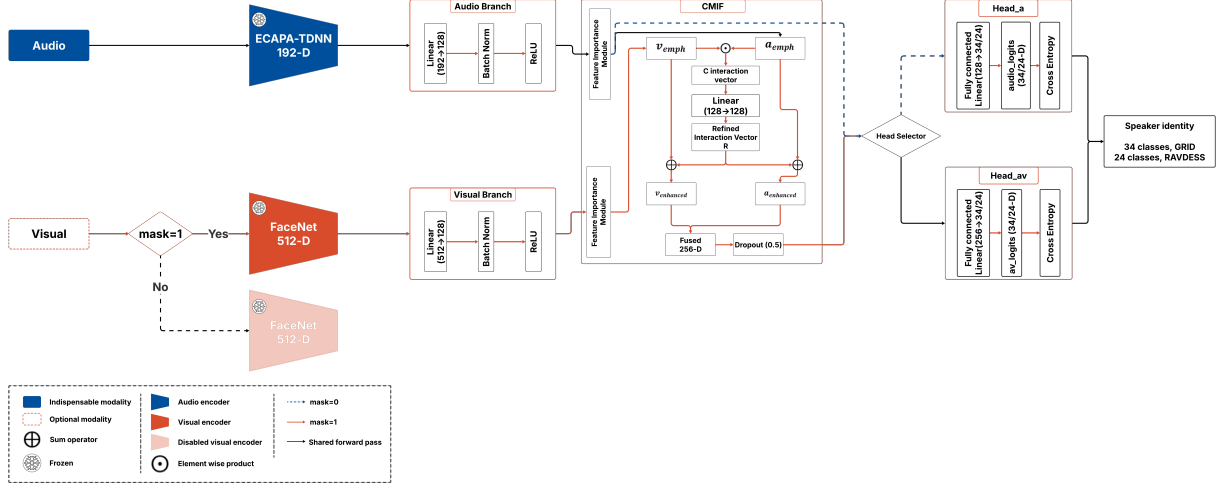


Figure 3: **RMMF on the CoMISI backbone.** Audio is the indispensable modality and the visual stream is optional, gated by a per-sample mask. When both modalities are present the CMIF module re-weights each stream with a feature importance module and exchanges cross-modal information before fusion. When the visual stream is absent a dedicated audio-only head is used. Both heads output speaker logits trained with cross-entropy.

Table 3: Speaker identification accuracy on GRID and RAVDESS. V: visual modality present, $\neg V$: visual modality absent. CoMISI (re-run backbone) and RMMF-V are averaged over 10 seeds. $\Delta(V/\neg V)$: RMMF-V gain over the CoMISI backbone (pp) / change of the audio-only $\neg V$ head relative to RMMF-V (pp). Audio-only: a standalone classifier on frozen ECAPA-TDNN embeddings (10 seeds).

Setting	CoMISI [4]	RMMF (V)	RMMF ($\neg V$)	Audio-only	$\Delta(V/\neg V)$
GRID – Clean	0.9788	0.9996	0.9880	0.9963	+2.08 / -1.16
GRID – Noisy	0.9997	0.9998	0.9740	0.9987	+0.01 / -2.58
RAVDESS – Clean	1.0000	1.0000	0.9510	0.5060	+0.00 / -4.90
RAVDESS – Noisy	0.9998	1.0000	0.9370	0.5162	+0.02 / -6.30

there. ECAPA-TDNN is trained for speaker verification on neutral speech, and under the strong emotional variation in RAVDESS its frozen embeddings track expressed emotion more than speaker identity. RMMF overcomes this by fine-tuning the audio encoder within the shared model, which is what lets the audio-only head recover the identity information (95.1%) that the frozen baseline cannot.²

5.3. Application 3: Cross-Lingual Multimodal Speaker Identification

Our full challenge submission achieves **1st place** on the POLY-SIM 2026 evaluation server. Table 4 isolates RMMF from the additional challenge-system components. On its own (raw Masked Fusion and Orthogonal Projection (MaskedFOP), no seed-averaging or label propagation), RMMF raises overall P-accuracy from 0.734 to 0.962

²The attribution to emotional variation follows from ECAPA-TDNN’s neutral-speech training domain and the separability statistics reported here, the released feature files retain only speaker labels, so emotion-conditioned clustering is inferred rather than measured directly.

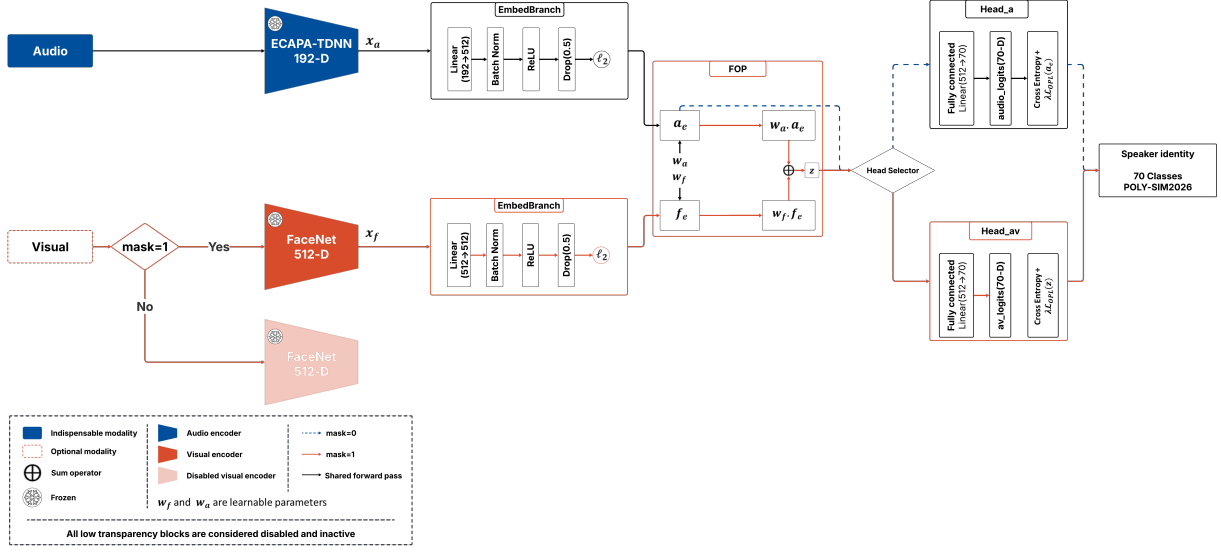


Figure 4: **RMMF on the FOP backbone (MaskedFOP)**. Audio is the indispensable modality and is never masked, while the face stream is optional and dropped per sample with probability 0.5. Each stream passes through an embed branch that produces an ℓ_2 -normalized embedding, and a linear fusion combines them into a joint representation. The mask selects one of two heads, a multimodal head when the face is present and an audio-only head when it is absent, and both heads are trained with cross-entropy together with an orthogonal projection loss. The audio-only head is supervised in both regimes, which keeps the audio branch discriminative when the face is missing.

(+0.228, mean over two seeds, ± 0.005). The gains are concentrated exactly where the missing-modality design should help: the audio-only settings P4 (English, $0.525 \rightarrow 0.984$) and P6 (Urdu, $0.439 \rightarrow 0.882$), where the face modality is absent at test time, while on the multimodal settings P3 and P5 (both modalities present, FOP already near ceiling) RMMF is at parity. This confirms the improvement comes from RMMF’s audio-only head $H_{(0)}$ and modality-dropout training rather than language-specific tuning, RMMF alone accounts for +0.444 of the P6 gain. Adding domain-adversarial training and two-stage label propagation yields our full **1st-place** challenge entry at 0.999 overall.

5.4. Ablation Studies

Table 5 isolates each RMMF mechanism on the FOP backbone, using the raw architecture on the local labeled validation split with no seed-averaging and no label propagation (seeds 1–6). The mask-gated dual head drives the missing-modality result. Removing interaction gating, so that audio-only inputs pass through the fused head, costs 5.4 pp on the cross-lingual audio-only setting P6 (0.866 to 0.812) and roughly doubles its variance (± 0.010 to ± 0.029), the drop holds across all six seeds (Wilcoxon $p = 0.031$). Collapsing the two heads into one shared classifier costs 1.5 pp on the cross-lingual multimodal setting P5 (0.968 to 0.954 , $p = 0.031$), which is where separate heads help. Modality dropout changes no setting significantly, because the audio head is already supervised through the fused branch at every step. Against the FOP baseline the improvement sits almost entirely in the audio-only regime: on P6 it is significant at the sample level (McNemar $p < 10^{-83}$, pooled over seeds), and it comes from RMMF preventing

Table 4: POLY-SIM 2026 P-accuracy. RMMF on the FOP backbone (MaskedFOP) raises overall P-accuracy from 0.734 to 0.962 (+0.228), with gains concentrated in the audio-only settings P4 (+0.459) and P6 (+0.444) where the face is absent. Domain-adversarial training and label propagation add the remaining margin to the full 1st-place system (0.999). The full-system row is transductive: its label propagation uses the unlabeled test set, so it is not comparable like-for-like with the inductive rows above.

Model	Overall	P3	P4	P5	P6
FOP (baseline) [40]	0.734	0.988	0.525	0.983	0.439
RMMF (MaskedFOP, ours) [†]	0.962	1.000	0.984	0.980	0.882
+ DA + label propagation (full system) [‡]	0.999	0.998	0.998	1.000	0.999

[†] Mean over 2 seeds, overall 0.962 ± 0.005 . Raw MaskedFOP, no seed-averaging or label propagation.

[‡] Transductive: label propagation accesses the unlabeled test set (not like-for-like with the inductive rows).

the baseline’s cross-lingual collapse, where FOP falls to 48.7% on one seed, rather than from a uniform margin. Six seeds is the smallest number at which a two-sided Wilcoxon signed-rank test can reach $p < 0.05$, since its minimum attainable value is $2/2^6 = 0.031$.

Table 5: Component ablation on the POLY-SIM local labeled validation split (raw MaskedFOP, no seed-averaging or label propagation, mean $\pm$ std over seeds 1–6). These local-validation numbers are not comparable to the challenge-test scores in Table 4.

Variant	P3	P4	P5	P6	Overall
FOP baseline	1.000	0.970 $\pm$ 0.031	0.977 $\pm$ 0.006	0.771 $\pm$ 0.151	0.930
RMMF (full)	1.000	0.987 $\pm$ 0.007	0.968 $\pm$ 0.007	0.866$\pm$0.010	0.955
– modality dropout	1.000	0.986 $\pm$ 0.006	0.972 $\pm$ 0.008	0.864 $\pm$ 0.015	0.956
– interaction gating	1.000	0.981 $\pm$ 0.009	0.968 $\pm$ 0.007	0.812 $\pm$ 0.029	0.940
– specialized heads (shared)	0.998 $\pm$ 0.002	0.987 $\pm$ 0.007	0.954 $\pm$ 0.007	0.877 $\pm$ 0.010	0.954

5.5. Discussion

- **Full-modality gains** - In all three use cases, RMMF improves performance over the original backbone even when all modalities are present. This is a non-trivial finding, since one might expect the additional complexity of specialized heads and modality dropout to slow convergence. Instead, modality dropout acts as a regularizer on the shared encoder, discouraging over-reliance on any single modality and producing more generalizable features.
- **Graceful degradation** - Missing-modality heads generally match or beat the corresponding unimodal baselines from the literature. This shows that the specialized heads benefit from the richer shared representations produced by full-modality training, rather than training from scratch on incomplete data. In practical terms, this means deploying RMMF is beneficial even for users who know in advance that a modality will be unavailable.

- **Indispensable modality design** - Designating the primary signal source as indispensable (WSI in pathology, audio in speaker recognition) reflects domain knowledge and keeps the head count manageable. An incorrect choice would require covering the absence of the most informative modality, likely producing substantially larger degradation and requiring far more missing-modality training samples to compensate.

6. Limitations and Broader Impact

Limitations. (i) RMMF assumes a genuinely indispensable modality exists and is correctly identified, tasks with no such modality fall back to the full $2^K - 1$ head count. (ii) The head count grows exponentially in the number of optional modalities, so RMMF targets small K (here $K \leq 3$). (iii) Specialized heads split training data per pattern, very rare patterns may be under-trained despite dropout. (iv) The head-parameter overhead, though small in absolute terms (a few $\times 10^4$ weights), is a larger fraction of compact backbones, up to 8.3% for FOP, Table 1 reports the measured overhead and wall-clock saving per backbone. **Broader impact.** Missing-modality robustness is most valuable in exactly the high-stakes settings (clinical prognosis, biometric identification) where it can also cause harm: a system that degrades gracefully may mask that it is operating on reduced evidence. Deployments should surface which modalities were present per prediction and calibrate confidence accordingly, RMMF’s explicit mask makes this auditing natural.

7. Reproducibility Statement

Data. All three applications are built on publicly available datasets, listed under Data availability, so that every result can be reproduced without access to private cohorts. In Application 1 the whole-slide images are tiled and encoded into patch features with the CLAM pipeline using the PLIP foundation model, the genomic profiles are expressed as pathway-level expression vectors, and the matched pathology reports are taken from the TCGA-Reports collection. Application 2 applies the audio-visual augmentation parameters described in Section 4. Application 3 uses the official POLY-SIM 2026 challenge partition.

Splits and optimization. Each application follows the evaluation protocol of its source benchmark, and the backbones are trained with their original optimizers so that the reported gains are attributable to RMMF rather than to retuning. The per-application splits, optimizers, and learning-rate schedules are reported with the experimental settings in Section 4.

8. Conclusion

We presented RMMF, a lightweight, backbone-agnostic framework that handles inference-time modality availability explicitly through three coordinated mechanisms: a binary mask that gates cross-modal interaction, specialized per-pattern heads that decouple decision boundaries across availability patterns, and modality dropout that supplies

gradient to every head while regularizing the shared encoder. A single indispensable-modality design principle keeps the head count at 2^{K-1} and the parameter overhead modest (0.24%–8.3%, Table 1). Instantiated unchanged on three different backbones across two unrelated domains, cancer survival prediction and (cross-lingual) audio-visual speaker identification, including a 1st-place POLY-SIM 2026 result, RMMF (i) matches or improves the full-modality result of the original backbone, with statistically significant gains on all three survival cohorts, and (ii) degrades gracefully when modalities are absent, matching or beating unimodal baselines in the missing-modality regime. The recurring full-modality gain reframes modality dropout from a robustness patch into a beneficial regularizer, and the cross-domain transfer indicates the principle is not tied to any one architecture or field. Future work includes scaling beyond small K via head factorization, learning rather than designating the indispensable modality, and calibrated confidence under partial evidence.

CRedit authorship contribution statement

Ayoub Elkhouchari: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing original draft, Visualization. **Youssef Iraqi:** Conceptualization, Edits, Supervision & Review. **Loubna Mekouar:** Conceptualization, Edits, Supervision & Review.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Declaration of generative AI in the writing process

The free-text pathology reports used in Application 1 were structured with a large language model as part of the data-processing pipeline, this methodological use is described in Section 4. No generative AI tool was used to write or generate the scientific content of this manuscript. The authors reviewed all material and take full responsibility for the content of the publication.

Data availability

All datasets used in this study are publicly available. Application 1 uses the TCGA cohorts (BLCA, BRCA, LUAD) [36], Application 2 uses the GRID and RAVDESS corpora, and Application 3 uses the MAV-Celeb dataset [41] together with the POLY-SIM 2026 challenge data. The RMMF wrappers, the single-head baseline variant used in the ablations, and the availability-pattern evaluation code will be available upon paper acceptance at <https://github.com/Ayoub-Elkhouchari/RMMF>.

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